INCH-POUND

A-A-58001
1 JUNE 1992

COMMERCIAL ITEM DESCRIPTION
GENERATOR, SIGNAL, DIGITAL WORD

The General Services Administration has authorized the use of this Commercial Item Description.

Abstract. This Description (CID) describes a Digital Signal Generator. This CID is meant as a minimum requirement for the Digital Signal Generator in which only those manufacturers that meet or surpass the following requirements are qualified per this CID.

Classification Equipment covered by this CID shall be commercially a available equipment and may be modified to the extent necessary to meet the following description.

MIL-T-28800. The equipment shall meet the following paragraphs of MIL-T-28800.

Paragraph	Title
1.2.2	Class 5
1.2.4	Color R
3.6.5.1	Enclosure Form, Style E
3.7.4.1	Vibration, sinuosoidal, Class 5
3.8.3	EMI I Type III
3.9.3.1	Accessible Potentials Type III
3.9.3.1.2	Leakage Current Protection, Type III

Order of Precedence. In the event of a conflict between the text of this CID and the references cities herein (except for associated detail specifications, specification sheets or MS standards) the text of this CID shall take precedence. Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Salient Characteristics. The equipment shall be capable of operation and bic sample tested within the accuracies, limits, and specifications herein

Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-T-28800 for the classification as stated herein.

Electrical Power Sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 115/230 volts AC (+ 10%) at line frequencies of 45 Rz to 66 Rz. Power consumption shall not exceed 630 watts. Transient state conditions shall meet MIL-T-28800 requirements for the classification specified herein.

Beneficial comment (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed Technology & Industrial Services Division, SA-ALC/TIRDM, Kelly AFB, TX by using the Standardization Document Improvement Proposal (DD Form 1426) 78241-5609, appearing at the end of this document or by letter.

AMSC N/A FSC 6625

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Temperature Requirements.

a. Operating temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of 0 C to +50 C.

b. Nonoperating temperature. The equipment shall meet its performance and accuracy requirements in an nonoperating environment of -40 C to +65 C.

Mean Time Between Failures (MTBF). The equipment Mean Time Between failures (MTBF) shall be 2,000 hours. The lower test MTBF shall be 2,000 hours and the upper test MTBF shall be 4,000 hours.

Pre-conditioning requirement Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

Calibration and maintenance adjustment. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustment. These adjustment shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustable, wherever possible, shall be accessible without removal of the instrument case or modules The calibration or interval shall be a period of one year or greater based or an operating time of 2000 hours.

Performance characteristics. The Digital Signal Generator shall have at least eight parallel channels which meet the requirements listed below and unless otherwise specified, all vernier controls shall be in the calibrated position and all performance requirements shall apply after a 30 minute warm up period in a laboratory ambient operating temperature.

Two of the eight channels shall have a minimum data rat of 100 Mbits/S in the serial mode with an even number of bits.

All eight channels shall have a minimum data rate of 50 Mbits/S in the parallel mode

Electrical function specification. Unless otherwise specified the Digital Signal Generator shall meet the following minimum requirements:

ECL Drivers. The ECL Drivers shall drive to a logic "low" of less or equal to -1.60V and shall drive to a logic "high" of greater than or equal to -1.02V

Rise and Fall Times: The rise and fall times for inputs shall not exceed 3 nano seconds (ns).

Skew: The Skew between channels shall be less than or equal to 6 ns.

Internal Clock. The internal clock shall have a programmable period from 20 nsec to 999 usec and shall be derived from either the system clock or an external clock.

Digitally Generatad Internal Clock Range and Resolution. The pattern duration using the internal clock shall have the range and resolution as follows:

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Range	Resolution
20 ns - 980 ns	20 ns
10 us - 999 us	1 us
1 ms - 99.9 ms	100 us
.1 s - .999 s	1 ms

External clock: The external clock input shall have a frequency range from 8 Hz to 100 MHz.

External clock resolution: External clock resolution shall not deviate more than 1 percent of the range.

TTL output. The digital signal generator shall be capable of a 50 Mbits/S TTL output for all eight channels The logic low level for the output shall be less than or equal to 0.5V. The logic high shall be greater than or equal to 2.4V. The rise and fall time for the TTL output shall be better than 6 ns into an open circuit

Data capacity. The Digital Signal Generator shall have a minimum capacity of 1024 bits per channel for all eight parallel per channels.

System integration. The digital signal generator shall be system integration through an interface bus which shall comply with the requirements of MIL-T-28800 paragraph 3.3.4.

Devise interface. The digital signal generator shall include minimum of one probe/and per channel plus one probe/lead per channel plus one probe/lead which shall be used for external clock trigger, or enable. All probes/leads shall have a frequency response of 100 MHz or greater and shall have a retractable hook tip All probes or poc connectors shall interface to the Digital word generator using make type BNC female

Signal input and output connectors. Unless otherwise specified, all signal input and output connectors shall be Type BNC female.

Equipment dimensions The nominal dimensions per component for the Digital Signal Generator shoulc be approximately 7.75 inches height x 17 inches wide x 34 inches deep. It shall be rack mountable to comply with the requirements of MIL-T-28800 section 3.6.5.1.1.

Equipment weight. The weight in including all accessories (excluding manual) per component, shall not exceed 53 lbs.

Programablility: The period for each state shall be programmable and shall be inoependent of all other channels.

Operating documentation. Each unit shall be delivered with an Operators, Maintenance, and Parts manual which has been approved in accordance with TM-86-01 and MIL-M-7298.

QUALITY ASSURANCE PROVISIONS

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the inspection requirements specified

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herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections, examinations, or tests set forth in this description where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

Contractor certification. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met.

If a product is manufactured to metric dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements. In accordance with Section 23.043 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

PACKAGING

Preservation, Packaging, Labeling, and Marking Preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

NOTES.

MIL-T-28800, MIL-M-7298 and Federal Standard No. 376 are available from the Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.

The following product(s) are known to have previously met the requirements of this CID and need not resubmit bid samples or certification. All other manufacturers must meet the requirements of this CID and submit to bid testing and certification as required.

Manufacturer	Model	NSN
HP Interface Technology	8175A-C01 RS-680/OPT 680-001	6625-01-225-8193

Custodians:

Air Force - 99

Preparing Activity:

Air Force - 82

Project 6625-F881

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